MINEGUARD : AN INTELLIGENT MINING SAFETY HELMET WITH DUAL-ZONE HARDWARE ARCHITECTURE AND ADAPTIVE POWER MANAGEMENT

- **PROBLEM STATEMENT:INTELLIGENT MINER SAFETY VIA DUAL-ZONE EMBEDDED ARCHITECTURE**
- **Team ID : HSN|TN|00803**
- **STATE/UT:TAMIL NADU**



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PROBLEM – PROPOSED SOLUTION

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THE PROBLEM :

- Underground miners face toxic gas exposure, unsafe environmental conditions, and unexpected falls or impacts.
- Conventional safety helmets lack integrated real-time hazard monitoring and intelligent emergency alerts.

This Leads to:
Miners face delayed hazard detection, increased safety risks, and limited emergency response, resulting in potentially life-threatening working conditions.

OUR INNOVATION

Real-Time Hazard Detection
Monitors toxic gases and environmental conditions.

Intelligent Emergency Alerts
Provides instant alerts during hazardous conditions.



Fall & Impact Detection
Identifies sudden falls and impacts using IMU sensors.

Dual-Zone Power Architecture
Improves weight distribution by separating sensing and electronic units.

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TECHNICAL APPROACH

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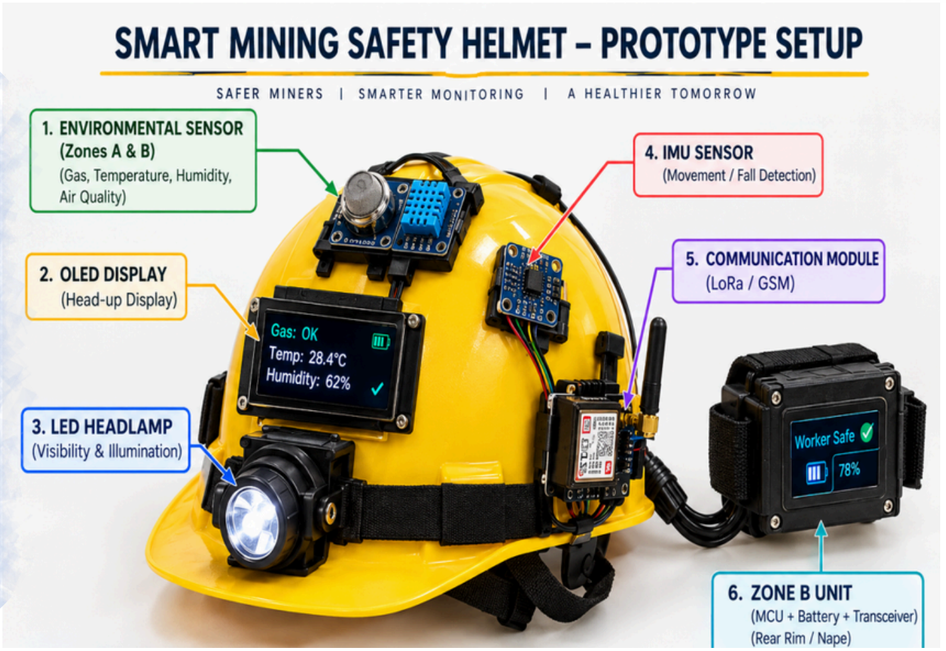
SOLUTION APPROACH :

***1. Sensor Integration*: ESP32 + Gas Sensors + IMU for hazard and fall detection.**

***2. Dual-Zone PCB Design*: KiCad/EasyEDA to design separate front and rear PCBs.**

***3. Smart Alerts & Power Saving*: Embedded C/C++ for emergency alerts and low-power operation.**

***4. Charging & Testing*: Inductive charging module + 3D helmet design + prototype validation.**



STAGE 1
Sensor Testing

→

STAGE 2
Dual-Zone PCB

→

STAGE 3
Firmware & Alerts

→

STAGE 4
Prototype Testing

- ***Development Flow***: *Sensor Integration → MCU Processing → Hazard Detection → Emergency Alert → Power-Efficient Operation*

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IMPACT POTENTIAL

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IMPACT

IMPROVED MINER SAFETY
Helps identify hazardous gases and unsafe environmental conditions at an early stage.

FASTER EMERGENCY RESPONSE
Alerts workers when dangerous conditions or detected emergencies require attention.

BETTER WORKING COMFORT
Aims to reduce front-heavy loading through balanced front and rear component placement.

SAFER MINING ENVIRONMENT
Supports hazard awareness and encourages safer working practices.



- Safety & Security
- Alert & Emergency
- Law & Regulation
- Engineering & Operations
- Monitoring & Analytics
- Sustainability & Clean Energy
- Connectivity & Smart Grid
- IoT & Smart Devices

BENEFITS

REAL-TIME MONITORING
Monitors selected gas, temperature, and worker-safety parameters.

POWER EFFICIENCY
Low-power operation helps reduce energy consumption and extend battery life.

CONTACTLESS CHARGING
Inductive charging eliminates exposed charging contacts.

COMPACT DUAL-ZONE DESIGN
Separates sensing components from control, power, and communication electronics.

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FEASIBILITY AND RISK ANALYSIS

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TECHNICAL FEASIBILITY
Available components
Dual-zone architecture
Modular development

FEASIBILITY

MITIGATION
Controlled testing
Power optimisation
Expert review and certification

ECONOMIC VIABILITY
₹14K–₹17K
Provisional prototype budget
Production costs to be assessed

VIABILITY

MARKET POTENTIAL
Potential users include mining companies and mine-safety organisations.
Addresses the need for hazard monitoring and worker safety.

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TEAM

TEAM NAME: VAYUJ NEXUS

TEAM ID:

TEAM LEAD - VIJAYA ABHIRAMI B K

- Project planning and team coordination
- System architecture and project management

DESIGN & INNOVATION HEAD - SHRUTHIKA R

- Dual-zone helmet design and innovation
- Ergonomic design and 3D visualization

TECHNICAL DEVELOPER - GNANA VADIVU P

- ESP32 programming and sensor integration
- Wireless communication, power management, and prototype development

CHALLENGES

- Identifying Skills: Finding members with complementary technical and creative skills.
- Role Allocation: Assigning responsibilities based on individual strengths and interests.
- Coordination: Ensuring effective communication and collaboration among team members.
- Time Management: Balancing project development with academic commitments.

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REFERENCES

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IEEE RESEARCH PAPERS

- IoT based Smart Helmet for Automated and Multi-parametric Monitoring of Underground Miners' Health Hazards (2022)
  Link: https://ieeexplore.ieee.org/document/9845621/
- Power Efficient Intelligent Helmet for Coal Mining Security and Alerting (2023)
  Link: https://ieeexplore.ieee.org/document/10169507/
- AI-Based Safety Helmet for Mining Workers Using IoT Technology and ARM Cortex-M (2023)
  Link: https://ieeexplore.ieee.org/document/10190347/
- IoT-Enabled Safety Helmet for Coal Miners: Real-Time Monitoring, Data Collection, and Analysis (2024)
  Link: https://ieeexplore.ieee.org/document/11026450/
- Energy Neutral Activity Monitoring: Wearables Powered by Smart Inductive Charging Surfaces (2016)
  Link: https://ieeexplore.ieee.org/document/7732986/